

A First-Moment-Constrained Rayleigh Problem on the Half-Line

Davide Marchini*

Abstract

For $L > 0$, consider nonnegative probability densities f on the half-line satisfying $f(0) = 0$, $\int_0^\infty f = 1$, and $\int_0^\infty xf(x) dx = L$. The Rayleigh quotient $\mathcal{R}(f) = \|f'\|_2^2 / \|f\|_2^2$ arises as the continuum fixed-risk turnover objective for a linear trend-following filter with serially independent volatility-adjusted returns. The infimum is attained by a unique density. Its continuous representative is positive on $(0, K_L)$, vanishes beyond K_L , and is linear plus trigonometric on its support. The free boundary is determined by a unique scalar root, giving $K_L/L \approx 2.40748835$ and $L^2\mathcal{R}(f_L) \approx 2.19907193$. The proof combines the direct method, a one-dimensional obstacle system, compression of positivity components, and scalar free-boundary analysis. Discrete minima converge to the continuum minimum, and interpolated near-minimizers converge to the optimizer. For finitely supported kernels, further results identify distributional conditions under which the same Rayleigh objective governs expected turnover; a profilewise short-memory limit is also established.

1 Introduction

The design of a smooth trend-following signal can be viewed as a filter-design problem. In a white-noise benchmark, fixing the risk of the signal and penalizing changes in the position reduces the problem to a ratio of quadratic energies: the numerator measures the squared first difference of the kernel, while the denominator fixes signal variance. A further scale constraint is needed to prevent the trivial solution obtained by making the filter arbitrarily slow. The first moment of the kernel itself provides the natural lookback constraint.

The main result establishes existence and uniqueness and identifies the optimizer explicitly. The proof begins with the direct method, derives a one-dimensional obstacle system, and uses compression of positivity components to obtain a single compact support interval. The free-boundary equations then reduce the profile to a linear-plus-trigonometric form determined by a unique scalar root. A discrete-to-continuum theorem gives convergence of the scaled minima and of interpolated near-minimizers. Further results treat finite-support distributional extensions and the profilewise short-memory limit under their stated hypotheses.

The statements proved here are accompanied by a Lean 4 formalization with mathlib, described in Appendix D. In particular the main theorem, the discrete convergence theorem, and the decimal constants printed above are machine-checked, and the certified arithmetic of Appendix C is reproduced by a script shipped with the formalization.

A closely related variational formulation was studied by Martin [1]. In his Section 2.2, path length is minimized under a unit L^2 norm and the constraint $\int_0^\infty t^{-1}K(t)^2 dt = 1/\tau$; EMA2= solves the resulting Euler–Lagrange equation and is optimal in that formulation. The present problem instead imposes mass, nonnegativity, zero trace, and first moment on

*Corresponding author: davide.marchini2@gmail.com

the kernel density. These different constraints lead to different Euler–Lagrange and free-boundary structures: the resulting optimizer is compactly supported, whereas the EMA2= kernel has infinite support and exponential decay.

Broad empirical work on time-series momentum and trend following provides economic context [7, 8], but does not address this kernel-design theorem. Related transaction-filter and dynamic portfolio work [5, 6] optimizes returns or trading policies under predictability and costs; EMA performance studies [3, 4] likewise differ from the fixed-risk, fixed-lookback minimum-turnover problem. This distinction also separates the present result from stochastic-control work such as Martin [2].

Under a common first-moment normalization, the two objective values have ratio approximately 0.5498; the corresponding Gaussian expected-turnover ratio is approximately 0.7415. The exact calculation is given in Section 10.

The remaining sections prove the theorem, establish discrete convergence, and record the distributional and serial-dependence qualifications of the extensions. Appendices A–B contain the corresponding proofs, and Appendices C–D record the certified arithmetic and the Lean 4 formalization.

2 The variational problem

For $L > 0$, define

$$\mathcal{A}_L = \left\{ f \in H^1(0, \infty) : f \geq 0 \text{ a.e., } f(0) = 0, \int_0^\infty f(x) dx = 1, \int_0^\infty x f(x) dx = L \right\}.$$

The value $f(0)$ is understood in the Sobolev trace sense. We study

$$\Lambda_L = \inf_{f \in \mathcal{A}_L} \mathcal{R}(f), \quad \mathcal{R}(f) = \frac{\|f'\|_{L^2(0, \infty)}^2}{\|f\|_{L^2(0, \infty)}^2}.$$

For $0 < t < 2\pi$, set

$$p(t) = 2 - t \cot \frac{t}{2}, \quad Q(t) = t^4 - 6t^2 p(t) + 3p(t)^3 (p(t) - 2).$$

As proved in Lemma 7.1, Q has a unique zero t_* in $(0, 2\pi)$. Define

$$A_* := \frac{t_* \sin t_* + \cos t_* - 1}{t_*(\cos t_* - 1)}, \quad C_* := \frac{t_* \cos t_* - \sin t_*}{t_*(\cos t_* - 1)}.$$

Theorem 2.1 (Existence, uniqueness, and characterization of the optimizer). *For every $L > 0$, the infimum Λ_L is attained by a unique density up to equality almost everywhere. Its continuous representative has positivity set $(0, K_L)$ and has the form*

$$f_L(x) = \begin{cases} \frac{1}{K_L I_0} F_*\left(\frac{x}{K_L}\right), & 0 \leq x \leq K_L, \\ 0, & x > K_L, \end{cases}$$

where

$$F_*(y) = A_* \sin(t_* y) + C_*(\cos(t_* y) - 1) + y,$$

with

$$t_* \approx 3.570129027812636, \quad A_* \approx 0.497710545413385, \quad C_* \approx 0.415370649651761.$$

Writing

$$I_0 = \int_0^1 F_*(y) dy, \quad I_1 = \int_0^1 y F_*(y) dy,$$

one obtains

$$\kappa_* := \frac{K_L}{L} = \frac{I_0}{I_1} = \frac{1}{C_*} = \frac{t_*(\cos t_* - 1)}{t_* \cos t_* - \sin t_*} \approx 2.407488350075723,$$

and

$$\mathcal{R}(f_L) = \frac{t_*^2}{K_L^2} = \frac{t_*^2 C_*^2}{L^2} \approx \frac{2.199071934562496}{L^2}.$$

In particular, the dimensionless minimum is

$$\lambda_* := L^2 \Lambda_L = t_*^2 C_*^2 \approx 2.199071934562496.$$

Equivalently, every $f \in \mathcal{A}_L$ satisfies

$$\int_0^\infty |f'|^2 dx \geq \frac{t_*^2 C_*^2}{L^2} \int_0^\infty f^2 dx,$$

with equality if and only if $f = f_L$ almost everywhere.

Remark 2.2 (Proof roadmap and analytic background). The proof is self-contained apart from standard one-dimensional Sobolev compactness, traces, weak lower semicontinuity, and direct-method facts [9, 19]. The multiplier measure, one-dimensional obstacle regularity, and compression argument are developed explicitly; the obstacle framework is classical [18, 10], with one-dimensional regularity as in Mandallena [12]. We also use the standard concavity, level-set, zero-extension, area, and trapezoidal facts needed in the proofs. Sharp Rayleigh and Poincaré context is surveyed by Kuznetsov and Nazarov [13]; related compactly supported Nash extremals and variational treatments are given by Carlen–Loss [16] and Bouin–Dolbeault–Schmeiser [17].

Remark 2.3 (The universal constant). The number κ_* is not introduced as a fitted numerical constant. It is the unique dimensionless free-boundary/eigenvalue constant generated by the optimization problem. Equivalently, if $x_* = t_*/2$, then

$$\kappa_* = \frac{2x_* \tan^2 x_*}{x_* \tan^2 x_* + \tan x_* - x_*},$$

where x_* is the unique admissible root inherited from the scalar consistency equation $Q(t) = 0$. We do not require, and do not know, an elementary closed-form representation of t_* or κ_* . The intrinsic root characterization is sufficient for the results below.

The uniqueness and numerical statements in Theorem 2.1 are proved in Sections 3–8 below; the theorem is stated here in advance so that its constants and equality case are available for reference.

3 Scaling

By Lemma 3.1 the problem at lookback L reduces to the problem at unit mean up to the explicit factor L^{-2} , so the unit-mean case carries all of the content.

Lemma 3.1. *If $g(y) = Lf(Ly)$, then $f \in \mathcal{A}_L$ if and only if $g \in \mathcal{A}_1$, and*

$$\mathcal{R}(f) = L^{-2} \mathcal{R}(g).$$

Thus $\Lambda_L = L^{-2} \Lambda_1$, and any optimal support endpoint must be proportional to L .

Proof. Substitute $x = Ly$ and $f(x) = L^{-1}g(x/L)$. Then

$$\int g = \int f = 1, \quad \int yg(y) dy = L^{-1} \int xf(x) dx = 1.$$

Moreover,

$$f'(x) = L^{-2}g'(x/L),$$

so

$$\|f'\|_2^2 = L^{-3}\|g'\|_2^2, \quad \|f\|_2^2 = L^{-1}\|g\|_2^2.$$

Taking the quotient proves the claim. $\square$

4 Compactness issues

Lemma 4.1 (Nondegeneracy of the denominator). *Every $f \in \mathcal{A}_L$ satisfies*

$$\|f\|_2^2 \geq \frac{1}{8L}.$$

Proof. The mean constraint gives

$$\int_{2L}^{\infty} f(x) dx \leq \frac{1}{2L} \int_0^{\infty} xf(x) dx = \frac{1}{2}.$$

Hence $\int_0^{2L} f \geq 1/2$. Cauchy–Schwarz gives

$$\frac{1}{4} \leq \left(\int_0^{2L} f \right)^2 \leq 2L \int_0^{2L} f^2 \leq 2L \|f\|_2^2.$$

$\square$

Lemma 4.2 (Uniform tail control). *Let (f_n) be a nonnegative sequence in $H^1(0, \infty)$ such that $\sup_n \|f'_n\|_2 < \infty$ and $\sup_n \int_R^{\infty} f_n(x) dx \leq \eta(R)$, where $\eta(R) \rightarrow 0$ as $R \rightarrow \infty$. Then*

$$\sup_n \int_R^{\infty} f_n(x)^2 dx \rightarrow 0 \quad (R \rightarrow \infty).$$

Proof. For fixed R , apply the one-dimensional Gagliardo–Nirenberg inequality on $(0, \infty)$ to the translated function $v_n(y) = f_n(R + y)$. Since $\|v_n\|_1 = \int_R^{\infty} f_n$ and $\|v'_n\|_2 \leq \|f'_n\|_2$,

$$\int_R^{\infty} f_n^2 \leq 2^{2/3} \left(\int_R^{\infty} f_n \right)^{4/3} \|f'_n\|_2^{2/3}.$$

The right-hand side tends to zero uniformly in n as $R \rightarrow \infty$. $\square$

Proposition 4.3 (Existence). *A minimizing sequence is bounded in $H^1(0, \infty)$, is locally precompact in L^2 and in the uniform topology, cannot lose probability mass at infinity, and cannot lose first moment at infinity without contradicting scaling. Consequently the direct method yields a minimizer.*

Proof. The class $\mathcal{A}_L$ is nonempty; for example, $6x(2L - x)_+ / (2L)^3$ belongs to it. Let $(f_n) \subset \mathcal{A}_L$ be minimizing and choose $Q < \infty$ such that $\mathcal{R}(f_n) \leq Q$. On any half-line, the one-dimensional Gagliardo–Nirenberg inequality gives

$$\|v\|_2^2 \leq 2^{2/3} \|v\|_1^{4/3} \|v'\|_2^{2/3}. \quad (1)$$

Indeed, $\|v\|_2^2 \leq \|v\|_1 \|v\|_\infty$ and $\|v\|_\infty^2 \leq 2\|v\|_2 \|v'\|_2$. Since $\|f_n\|_1 = 1$, writing $X_n = \|f'_n\|_2$ yields

$$X_n^2 \leq Q \|f_n\|_2^2 \leq 2^{2/3} Q X_n^{2/3}.$$

Thus (f_n) is bounded in $H^1(0, \infty)$. After taking a subsequence, $f_n \rightharpoonup f$ in $H^1(0, \infty)$. On every bounded interval the compact embedding $H^1 \hookrightarrow C^0$ in one dimension, followed by a diagonal argument, gives local uniform and local L^2 convergence. In particular, $f \geq 0$ and continuity of the trace gives $f(0) = 0$.

The first moment gives tightness of probability mass:

$$\int_R^\infty f_n(x) dx \leq \frac{L}{R}.$$

Together with local convergence, this proves $\int f = 1$. Lemma 4.2, together with

$$\int_R^\infty f_n(x) dx \leq \frac{L}{R},$$

shows that the L^2 tails vanish uniformly. By Fatou's lemma, the same tail bound holds for the limit. Combining it with strong local L^2 convergence proves $f_n \rightarrow f$ strongly in $L^2(0, \infty)$. Fatou's lemma also gives

$$m := \int_0^\infty x f(x) dx \leq L.$$

We have $m > 0$: if $m = 0$ then $x f(x) = 0$ almost everywhere on $(0, \infty)$, so $f = 0$ almost everywhere there, contradicting $\int f = 1$. Weak lower semicontinuity of the Dirichlet integral and strong convergence of the denominator imply

$$\mathcal{R}(f) \leq \liminf_{n \rightarrow \infty} \mathcal{R}(f_n) = \Lambda_L.$$

The only remaining possible defect is that a vanishing amount of mass carries nonzero first moment to infinity. If $m < L$, dilate the limit to mean L by

$$\tilde{f}(x) = \frac{m}{L} f\left(\frac{m}{L}x\right)$$

to obtain an admissible density satisfying

$$\mathcal{R}(\tilde{f}) = \left(\frac{m}{L}\right)^2 \mathcal{R}(f) < \mathcal{R}(f).$$

The quotient $\mathcal{R}(f)$ is positive: otherwise $f' = 0$ almost everywhere, so f would be a constant L^2 function and hence would vanish, contrary to $\int f = 1$. This is the nondegeneracy of Lemma 4.1 in the case $L = 1$. Thus the last display is strictly below Λ_L , a contradiction. Therefore $m = L$, $f \in \mathcal{A}_L$, and f is a minimizer. $\square$

5 Euler–Lagrange and obstacle system

Let f be a minimizer and write

$$q = \mathcal{R}(f) > 0.$$

Indeed, $f \not\equiv 0$ and $f(0) = 0$, so $\int_0^\infty |f'|^2 > 0$.

Proposition 5.1 (Optimality system and local regularity). *There are constants a, b and a nonnegative Radon measure μ , supported on the contact set $\{f = 0\}$, such that*

$$-f'' - qf + a + bx = \mu \quad (2)$$

in distributions, with

$$f \geq 0, \quad \mu \geq 0, \quad \int f \, d\mu = 0.$$

Moreover, $\mu \in L_{\text{loc}}^\infty(0, \infty)$ and $f \in W_{\text{loc}}^{2,\infty}(0, \infty) \subset C_{\text{loc}}^{1,1}(0, \infty)$.

Proof. Set

$$\ell(h) := \int_0^\infty f' h' \, dx - q \int_0^\infty f h \, dx, \quad M(h) := \left(\int_0^\infty h \, dx, \int_0^\infty x h \, dx \right).$$

The derivative of the quotient at f is

$$D\mathcal{R}(f)[h] = \frac{2}{\|f\|_2^2} \ell(h).$$

The positivity set $\Omega = \{f > 0\}$ is a nonempty open set. Choose two nonnegative functions $\phi_1, \phi_2 \in C_c^\infty(\Omega)$ whose supports lie in small disjoint neighborhoods of two distinct points of Ω . By shrinking the supports if necessary, their mass–moment vectors

$$M(\phi_i) = \left(\int_0^\infty \phi_i \, dx, \int_0^\infty x \phi_i \, dx \right)$$

are linearly independent: for a bump concentrated near x_i , the ratio of its first moment to its mass is arbitrarily close to x_i . Let $R : \mathbb{R}^2 \rightarrow \text{span}\{\phi_1, \phi_2\}$ be the resulting linear right inverse of M .

For a nonnegative test function $\psi \in C_c^\infty(0, \infty)$, put

$$h_\psi = \psi - R(M\psi).$$

Then $M(h_\psi) = 0$. Outside the fixed compact support of the ϕ_i , the negative part of h_ψ vanishes; on that support, f has a positive minimum. Consequently $f + \varepsilon h_\psi \geq 0$ for all sufficiently small $\varepsilon > 0$. This is an admissible one-sided variation, so minimality gives $\ell(h_\psi) \geq 0$.

The linear functional $\ell \circ R$ on $\mathbb{R}^2$ has the form $(u, v) \mapsto -au - bv$ for unique constants a, b . Therefore

$$\mu(\psi) := \ell(\psi) + a \int \psi + b \int x\psi \geq 0 \quad (\psi \in C_c^\infty, \psi \geq 0).$$

Thus μ is a positive distribution and hence, by the standard theorem on positive distributions, a locally finite nonnegative Radon measure. If ψ is supported compactly in Ω , then both signs of h_ψ are admissible for small parameter, so $\mu(\psi) = 0$. Hence the restriction of the positive measure μ to every compact subset of Ω is zero, and therefore $\text{supp } \mu \subseteq \{f = 0\}$, and the definition of ℓ gives (2). Since f is continuous and vanishes on the support of μ , $f = 0$ μ -almost everywhere, which is the stated complementarity relation.

It remains to prove the asserted regularity. We use the following elementary one-dimensional fact. If J is a bounded interval, $v \in H^1(J)$, $\psi \in C^{1,1}(J)$, $v \geq \psi$, and the distribution $-v''$ is a nonnegative measure supported on $\{v = \psi\}$, then $v \in C_{\text{loc}}^{1,1}(J)$. Here and below we use the continuous representative of v . The distributional inequality $-v'' \geq 0$

implies that this representative is concave: equivalently, v' has a nonincreasing bounded-variation representative, and integration recovers the concavity inequality for v . In particular the one-sided derivatives D^-v and D^+v exist at every interior point and satisfy $D^-v \geq D^+v$.

At an interior contact point x , the inequalities $v(x+h) - v(x) \geq \psi(x+h) - \psi(x)$ for positive and negative h give

$$D^+v(x) \geq \psi'(x) \geq D^-v(x).$$

Thus all three quantities are equal, so v is differentiable at every contact point and $v'(x) = \psi'(x)$. On each component of the open set $\{v > \psi\}$ the measure $-v''$ vanishes, so v is affine there. It follows that v is differentiable at every interior point of J unless the whole interval is one free component; in that exceptional case v is affine and the conclusion is immediate.

It remains to bound the variation of its derivative. If two interior points $x < y$ lie in the same free component, then $v'(x) = v'(y)$. Otherwise, let r be x when x is a contact point and the right endpoint of the free component containing x otherwise. Define s from y analogously, using the left endpoint of its free component. The assumption that x and y do not lie in the same free component ensures that these endpoints are interior contact points and that $x \leq r \leq s \leq y$. Affineness on the free components and differentiability at their contact endpoints give $v'(x) = \psi'(r)$ and $v'(y) = \psi'(s)$. Concavity and the Lipschitz property of ψ' now give

$$0 \leq v'(x) - v'(y) = \psi'(r) - \psi'(s) \leq \|\psi'\|_{\text{Lip}}(s-r) \leq \|\psi'\|_{\text{Lip}}(y-x).$$

If the contact set is empty, the support assumption gives $-v'' = 0$ throughout J , and v is affine, so the same conclusion is immediate. This proves that v' is locally Lipschitz and hence that $v \in C_{\text{loc}}^{1,1}(J)$.

Now fix $J \Subset (0, \infty)$. Since one-dimensional H^1 functions are continuous, $F := qf - a - bx$ is bounded on J . Choose $h \in W^{2,\infty}(J)$ with $-h'' = F$ and put $v = f - h$. Equation (2) becomes

$$-v'' = \mu, \quad v \geq -h, \quad \text{supp } \mu \subseteq \{v = -h\}.$$

The preceding fact, with obstacle $-h$, gives $v \in C_{\text{loc}}^{1,1}(J)$ and hence $f \in W_{\text{loc}}^{2,\infty}(J)$. Finally (2) shows that μ has an L_{loc}^∞ density. $\square$

On every component of the positivity set, Proposition 5.1 gives

$$f''(x) + qf(x) = a + bx. \quad (3)$$

Lemma 5.2 (Contact-set compression). *Let $u \in H^1(0, \infty)$ be nonnegative, satisfy $u(0) = 0$, be integrable, and have finite first moment. Put $P = \{u > 0\}$ and*

$$T(x) = \int_0^x \mathbf{1}_P(s) ds.$$

There is a nonnegative $g \in H^1(0, \infty)$ with $g(0) = 0$ such that

$$\int g = \int u, \quad \int g^2 = \int u^2, \quad \int |g'|^2 = \int |u'|^2, \quad (4)$$

and

$$\int_0^\infty yg(y) dy = \int_0^\infty T(x)u(x) dx \leq \int_0^\infty xu(x) dx. \quad (5)$$

The last inequality is strict precisely when

$$\int_P |P^c \cap (0, x)| u(x) dx > 0,$$

where $|\cdot|$ denotes Lebesgue measure; in particular, this holds whenever the zero set has positive measure below a subset of P carrying positive $u(x) dx$ mass.

Proof. Use the continuous representative of u and write the open set P as the disjoint union of its countably many components $I_n = (a_n, b_n)$. The endpoint b_n is allowed to be infinite. The map T is absolutely continuous and satisfies $T' = \mathbf{1}_P$ almost everywhere. Consequently, on I_n ,

$$T(x) = T(a_n) + x - a_n,$$

so T maps I_n isometrically onto

$$\widehat{I}_n := (T(a_n), T(a_n) + b_n - a_n).$$

These image intervals have disjoint interiors. They cover $(0, |P|)$ up to a null set, where this interval is understood as $(0, \infty)$ if $|P| = \infty$. Indeed, the range of the continuous nondecreasing function T contains every y with $0 < y < |P|$. If such a y is not in any $\widehat{I}_n$, a point x with $T(x) = y$ must lie in P^c . On every bounded interval, the standard absolute-continuity estimate

$$|T(E)| \leq \int_E |T'(x)| dx$$

shows that $T(P^c)$ has measure zero, because $T' = 0$ almost everywhere on P^c . Exhausting the half-line by bounded intervals proves the asserted coverage. It follows that, for every nonnegative measurable φ , change of variables on the components gives the push-forward identity

$$\sum_n \int_{\widehat{I}_n} \varphi(y) dy = \int_P \varphi(T(x)) dx = \int_0^{|P|} \varphi(y) dy.$$

For each n , translate $u|_{I_n}$ to $\widehat{I}_n$ by $U_n(T(x)) = u(x)$ and extend it by zero outside $\widehat{I}_n$. At every finite endpoint of I_n , continuity gives $u = 0$, since the endpoint does not belong to P . If $b_n = \infty$, then $u(x) \rightarrow 0$ as $x \rightarrow \infty$, because a half-line H^1 function is uniformly continuous and belongs to L^2 . The standard one-dimensional zero-extension criterion therefore gives $U_n \in H^1(0, \infty)$, with

$$U'_n(T(x)) = u'(x) \quad \text{for almost every } x \in I_n.$$

The translated intervals are disjoint and every translated piece meeting the origin has zero trace there. Thus

$$g_N := \sum_{n=1}^N U_n$$

belongs to $H^1(0, \infty)$, is nonnegative, and satisfies $g_N(0) = 0$. For $M > N$, disjointness and translation invariance give

$$\|g_M - g_N\|_2^2 = \sum_{N < n \leq M} \int_{I_n} u^2, \quad \|g'_M - g'_N\|_2^2 = \sum_{N < n \leq M} \int_{I_n} |u'|^2.$$

An absolutely continuous function has derivative zero almost everywhere on each level set, so $u' = 0$ almost everywhere on $P^c = \{u = 0\}$. Hence the two series on the right exhaust the full square and Dirichlet energies of u , and (g_N) is Cauchy in $H^1(0, \infty)$. Let g be its limit. Continuity of the trace gives $g(0) = 0$, and convergence of the two norm identities gives

$$\int g^2 = \int u^2, \quad \int |g'|^2 = \int |u'|^2.$$

Because the U_n are nonnegative and have disjoint interiors, g_N increases pointwise to the function obtained by placing all translated pieces. Its pointwise limit agrees almost everywhere with the H^1 limit g , as follows by taking an almost-everywhere convergent subsequence

of the L^2 -convergent sequence. Monotone convergence, translation on each component, and the push-forward identity now give

$$\int_0^\infty g(y) dy = \sum_n \int_{I_n} u(x) dx = \int_0^\infty u(x) dx$$

and

$$\int_0^\infty yg(y) dy = \sum_n \int_{I_n} T(x)u(x) dx = \int_P T(x)u(x) dx = \int_0^\infty T(x)u(x) dx.$$

The last integral is finite because $0 \leq T(x) \leq x$. This proves all three identities in (4) and the equality in (5).

Finally,

$$x - T(x) = |P^c \cap (0, x)| \geq 0,$$

and therefore

$$\int_0^\infty xu(x) dx - \int_0^\infty T(x)u(x) dx = \int_P |P^c \cap (0, x)|u(x) dx.$$

This proves both the inequality and its stated strictness criterion. $\square$

Proposition 5.3 (Support geometry and smooth fit). *There is a finite $K > 0$ such that $f = 0$ on (K, ∞) , the obstacle reaction vanishes on $(0, K)$, and*

$$f(0) = 0, \quad f(K) = 0, \quad f'(K^-) = 0.$$

Consequently equation (3) holds throughout $(0, K)$. The scalar analysis below shows that in fact $f > 0$ on $(0, K)$.

Proof. By Proposition 5.1, $f \in C_{\text{loc}}^{1,1}$ and $\mu \in L_{\text{loc}}^\infty$. Apply Lemma 5.2 to f . The compressed function has the same mass, L^2 norm, and Dirichlet energy, hence the same Rayleigh quotient. Let m denote its first moment. We have $m > 0$: otherwise $yg(y) = 0$ almost everywhere, so the nonnegative function g would vanish almost everywhere on $(0, \infty)$, contradicting $\int g = 1$. If $m < L$, the dilation

$$\tilde{g}(x) = \frac{m}{L} g\left(\frac{m}{L}x\right)$$

belongs to $\mathcal{A}_L$ and satisfies $\mathcal{R}(\tilde{g}) = (m/L)^2 \mathcal{R}(f) < \mathcal{R}(f)$, contradicting minimality. Hence compression cannot strictly decrease the first moment. By the strictness statement in Lemma 5.2, this rules out both an initial zero interval followed by positive mass and any positive-measure gap before positive mass. Here is the measure argument explicitly. If the contact set had positive measure in the interior of the convex hull of $P = \{f > 0\}$, then there would be an R strictly below the right endpoint of the hull such that $E := P^c \cap (0, R)$ has positive measure. Since R lies below that endpoint, $P \cap (R, \infty)$ is nonempty; openness of P therefore supplies a nonempty open interval $J \subset P \cap (R, \infty)$. For every $x \in J$ one has $|P^c \cap (0, x)| \geq |E|$, and $\int_J f(x) dx > 0$. Hence

$$\int_P |P^c \cap (0, x)|f(x) dx \geq |E| \int_J f(x) dx > 0,$$

contradicting the absence of strict moment decrease. The same argument with E contained in an initial zero interval rules out a positive left endpoint of the convex hull. Thus the convex hull of $\{f > 0\}$ begins at the origin and the contact set has measure zero inside that

hull. Since $\mu \in L_{\text{loc}}^\infty$ and is supported on the contact set, it vanishes in the interior of the hull. If that hull were unbounded, equation (3) would hold on all of $(0, \infty)$ and would give

$$f(x) = A \sin(kx) + B \cos(kx) + \frac{a}{q} + \frac{b}{q}x, \quad k = \sqrt{q},$$

which cannot belong to $L^1 \cap L^2$ unless it vanishes identically. Indeed, $b \neq 0$ gives linear growth. If $b = 0$, then the remaining function is a constant plus a periodic trigonometric polynomial. Its squared mean over a period is

$$\frac{A^2 + B^2}{2} + \left(\frac{a}{q}\right)^2,$$

which is positive unless $A = B = a = 0$; in the nonzero case its L^2 integral on the half-line therefore diverges. Thus the hull is $[0, K]$ for some finite K , and equation (3) holds throughout $(0, K)$.

Continuity gives $f(K) = 0$. Since $f \in C_{\text{loc}}^{1,1}(0, \infty)$ and is identically zero immediately to the right of K , its derivative is continuous at the interior point K and $f'(K^-) = f'(K^+) = 0$. $\square$

6 Solution on the positive interval

Let $k = \sqrt{q}$. Solving (3) on $(0, K)$ gives

$$f(x) = A \sin(kx) + B \cos(kx) + \frac{a}{q} + \frac{b}{q}x.$$

The condition $f(0) = 0$ permits the form

$$f(x) = A \sin(kx) + C(\cos(kx) - 1) + Dx.$$

Here $C = -a/q$ and $D = b/q$. The right-hand side of (3) is bounded on $(0, K)$, so $f'' = a + bx - qf$ belongs to $L^\infty(0, K)$ and f extends to $W^{2,\infty}(0, K)$. Multiplying (3) by f , integrating over $(0, K)$, and using $f(0) = f(K) = 0$ therefore gives

$$-\int_0^K |f'|^2 + q \int_0^K f^2 = a \int_0^K f + b \int_0^K xf.$$

Using $q = \|f'\|_2^2 / \|f\|_2^2$, compact support, and the mass and first-moment constraints yields

$$a + bL = 0. \tag{6}$$

On (K, ∞) , the reaction measure has density $a + bx$, so its nonnegativity forces $b \geq 0$. If $b = 0$, then (6) gives $a = 0$, and the homogeneous equation together with $f(0) = f(K) = f'(K^-) = 0$ forces $f = 0$. Therefore $D > 0$.

Put

$$y = \frac{x}{K}, \quad t = kK.$$

After dividing the profile by the positive factor DK , define

$$F_t(y) := \frac{f(Ky)}{DK} = A_t \sin(ty) + C_t(\cos(ty) - 1) + y. \tag{7}$$

The multiplier identity therefore gives the useful relation

$$C_t = \frac{C}{DK} = -\frac{a}{bK} = \frac{L}{K}. \tag{8}$$

The free-boundary conditions are

$$F_t(1) = 0, \quad F'_t(1) = 0.$$

Solving the resulting two linear equations gives

$$A_t = \frac{t \sin t + \cos t - 1}{t(\cos t - 1)}, \quad (9)$$

and

$$C_t = \frac{t \cos t - \sin t}{t(\cos t - 1)}. \quad (10)$$

At $t = 2\pi n$, $n \geq 1$, the boundary equation $F_t(1) = 0$ is inconsistent, so these singular values do not produce profiles.

7 Rayleigh self-consistency

For a profile

$$f(x) = cF_t(x/K)\mathbf{1}_{[0,K]}(x),$$

a change of variables gives

$$\int_0^K f'(x)^2 dx = \frac{c^2}{K} \int_0^1 F'_t(y)^2 dy$$

and

$$\int_0^K f(x)^2 dx = c^2 K \int_0^1 F_t(y)^2 dy.$$

Therefore

$$\mathcal{R}(f) = \frac{1}{K^2} \frac{\int_0^1 F'_t(y)^2 dy}{\int_0^1 F_t(y)^2 dy}.$$

But $q = k^2 = t^2/K^2$. Hence t must solve

$$G(t) := \int_0^1 F'_t(y)^2 dy - t^2 \int_0^1 F_t(y)^2 dy = 0. \quad (11)$$

For the minimizing profile obtained above, $c = DK > 0$; thus F_t is nonnegative and nonzero on $[0, 1]$. There is a useful simplification. Directly from (7),

$$F_t'' + t^2 F_t = t^2(y - C_t).$$

Multiplication by F_t and integration by parts therefore show that, with

$$I_0(t) = \int_0^1 F_t(y) dy, \quad I_1(t) = \int_0^1 y F_t(y) dy,$$

one has

$$G(t) = t^2(C_t I_0(t) - I_1(t)). \quad (12)$$

For the minimizing profile, $I_0(t) > 0$ and the mass and moment constraints give $L/K = I_1(t)/I_0(t)$. Comparison with (8) therefore yields $C_t = I_1(t)/I_0(t)$, which is equivalent to $G(t) = 0$.

Lemma 7.1 (Positive branch and unique consistency root). *For every $0 < t < 2\pi$, the profile F_t is strictly positive on $(0, 1)$. No value $t > 2\pi$ for which the boundary system is defined gives a nonnegative profile. Moreover, G has exactly one zero in $(0, 2\pi)$.*

Proof. Let

$$\alpha(t) = \frac{t - \sin t}{1 - \cos t}.$$

Writing $s = t(1 - y)$ and using the two conditions at $y = 1$ gives the factorization

$$tF_t(y) = (\alpha(t) - \alpha(s))(1 - \cos s). \quad (13)$$

The derivative of α has the sign of

$$d(t) = (1 - \cos t)^2 - (t - \sin t) \sin t.$$

For $0 < t < \pi$, one has $d(0) = d'(0) = 0$ and $d''(t) = t \sin t > 0$; for $\pi \leq t < 2\pi$, positivity follows directly because $\sin t \leq 0$. Thus α is strictly increasing on $(0, 2\pi)$, and (13) proves $F_t > 0$ there. If $t > 2\pi$, choose $y = 1 - 2\pi/t$ in the equivalent, unfactored identity

$$tF_t(y) = \alpha(t)(1 - \cos s) - s + \sin s;$$

then $s = 2\pi$ and $tF_t(y) = -2\pi < 0$.

It remains to prove uniqueness of the root. Recall that $p(t) = F'_t(0) = 2 - t \cot(t/2)$ and $Q(t) = t^4 - 6t^2p(t) + 3p(t)^3(p(t) - 2)$. For reproducibility, direct integration gives

$$I_0(t) = \frac{t^2(1 + \cos t) - 4t \sin t + 4(1 - \cos t)}{2t^2(1 - \cos t)},$$

$$I_1(t) = \frac{t(\cos t + 2) - 3 \sin t}{6t(1 - \cos t)}.$$

Substitution in (12) simplifies to

$$G(t) = -\frac{Q(t)}{12t^2}. \quad (14)$$

For $0 < t \leq \pi$, the elementary inequality $x \cot x \leq 1 - x^2/3$ for $0 < x \leq \pi/2$ gives $p(t) \geq t^2/6$, while $p(t) \leq 2$. For completeness, after multiplying by $\sin x > 0$, the inequality is equivalent to

$$H(x) := (1 - x^2/3) \sin x - x \cos x \geq 0.$$

Here $H(0) = 0$ and $H'(x) = \frac{x}{3}(\sin x - x \cos x) > 0$ for $x > 0$, since $(\sin x - x \cos x)' = x \sin x > 0$. Thus the cotangent inequality is strict for $x > 0$, and hence $p(t) > t^2/6$. Therefore

$$t^4 - 6t^2p(t) < 0, \quad 3p(t)^3(p(t) - 2) \leq 0,$$

which proves $Q(t) < 0$ on this interval.

For $\pi < t \leq \sqrt{12}$, write

$$u = p - 2 = -t \cot \frac{t}{2} = t \tan \frac{t - \pi}{2}.$$

Here $0 < u < 3/5$: indeed, the elementary bounds $\pi > 3.14$ and $\sqrt{12} < 3.47$ give $(t - \pi)/2 < 1/6$, while $\sin x \leq x$ and $\cos x \geq 1 - x^2/2$ give $u < (7/2)(12/71) < 3/5$. Consequently

$$Q = t^2(t^2 - 12) + u(3u^3 + 18u^2 + 36u + 24 - 6t^2) < 0,$$

because the polynomial in the bracket before $-6t^2$ is at most $6591/125 < 54 < 6t^2$.

Finally, differentiation gives

$$p'(t) = \frac{t^2 + p^2 - 2p}{2t}.$$

For $\sqrt{12} \leq t < 2\pi$, substitution of $p = 2 + u$ yields

$$Q'(t) = \frac{1}{t} \left(t^2(t^2 - 12) + 6t^2u^3 + 24t^2u^2 + 18t^2u + 6u^5 + 39u^4 + 90u^3 + 84u^2 + 24u \right) > 0.$$

Since $p(t) = 2 - t \cot(t/2) \rightarrow \infty$ as $t \uparrow 2\pi$, the leading term $3p(t)^4$ shows that $Q(t) \rightarrow \infty$. Together with $Q(\sqrt{12}) < 0$ and the strict monotonicity above, this gives exactly one zero $t_* \in (\sqrt{12}, 2\pi)$. Equation (14) completes the proof. $\square$

The identities relating G and Q are exact algebraic identities, not numerical approximations. In particular, direct differentiation gives

$$p'(t) = \frac{t^2 + p(t)^2 - 2p(t)}{2t},$$

and substitution into Q' produces the displayed expression after writing $p = 2 + u$. Likewise, substituting the closed forms for A_t and C_t into G gives $G(t) = -Q(t)/(12t^2)$. The interval calculation below is therefore used only to certify the decimal representation of the unique root, not to establish uniqueness.

Directed-rounding interval evaluation gives

$$Q(3.570129027812636) < 0 < Q(3.570129027812637),$$

so

$$t_* \approx 3.570129027812636.$$

Equations (9) and (10) then give

$$A_* \approx 0.497710545413385, \quad C_* \approx 0.415370649651761.$$

Lemma 7.1 also completes the proof that every minimizer has positivity set $(0, K)$.

8 Mass, mean, and support

Define

$$I_0 = \int_0^1 F_*(y) dy, \quad I_1 = \int_0^1 y F_*(y) dy.$$

For

$$f_L(x) = c F_*\left(\frac{x}{K}\right) \mathbf{1}_{[0, K]}(x),$$

normalization gives

$$cK I_0 = 1,$$

and the mean condition gives

$$cK^2 I_1 = L.$$

Consequently,

$$c = \frac{1}{K I_0}, \quad K = L \frac{I_0}{I_1}.$$

By (12), $I_1/I_0 = C_*$, so the equivalent identity

$$\frac{K}{L} = \frac{1}{C_*} \quad (15)$$

provides an independent check on the support calculation. Numerically,

$$I_0 \approx 0.302496116691908, \quad I_1 \approx 0.125648008507453,$$

so

$$\frac{K}{L} \approx 2.407488350075723.$$

Finally,

$$\mathcal{R}(f_L) = \frac{t_*^2}{K^2} = \frac{1}{L^2} \frac{t_*^2}{(I_0/I_1)^2} \approx \frac{2.199071934562496}{L^2}.$$

Proposition 4.3 supplies a minimizer, and Proposition 5.3 supplies the support interval $[0, K]$ and the endpoint identities. The preceding argument shows that every minimizer has the same value t_* and hence the same dimensionless profile F_* . The mass and mean equations then determine c and K uniquely. This proves Theorem 2.1.

The optimizer has the scale-free representation

$$\Phi(z) := Lf_L(Lz) = \frac{1}{\kappa I_0} F_*\left(\frac{z}{\kappa}\right) \mathbf{1}_{[0, \kappa]}(z), \quad \kappa = \frac{K_L}{L}.$$

Thus Φ is independent of L , with $\int \Phi = 1$ and $\int z\Phi(z) dz = 1$. Figure 1 shows this universal kernel.

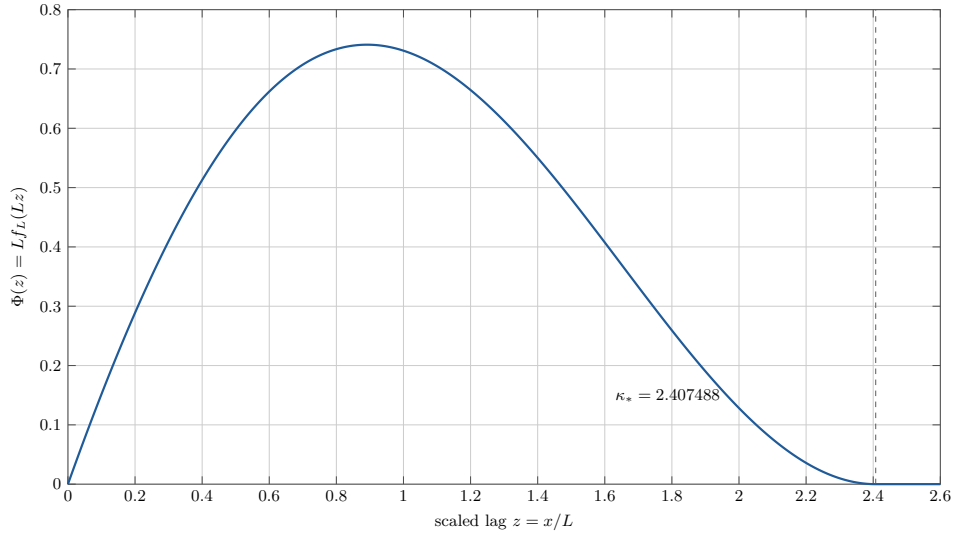


Figure 1: The universal optimal kernel in scaled coordinates. The curve is $\Phi(z) = Lf_L(Lz)$, and its support ends at $\kappa = K_L/L \approx 2.407488350075723$.

9 Endpoint behavior

At the origin,

$$F_*(0) = 0, \quad F'_*(0) = 1 + t_* A_* > 0, \quad F'_*(0) \approx 2.7768908656,$$

so

$$f_L(x) \sim c_0 x \quad (x \downarrow 0).$$

At the free boundary,

$$F_*(1) = F'_*(1) = 0, \quad F''_*(1) > 0, \quad F''_*(1) \approx 7.4515812118,$$

so Taylor expansion gives

$$f_L(x) \sim c_K (K_L - x)^2 \quad (x \uparrow K_L).$$

Thus the exact endpoint powers match those of a beta density proportional to $x^{2-1}(K_L - x)^{3-1}$, although the interior profile is not exactly beta.

10 Financial interpretation of the Rayleigh quotient

Let $(r_t)_{t \in \mathbb{Z}}$ be centered, jointly Gaussian, volatility-adjusted returns satisfying

$$\mathbb{E}[r_t r_s] = \delta_{ts}.$$

Thus the returns have unit variance and are serially uncorrelated; joint Gaussianity also makes them independent. For a nonzero square-summable sequence of lookback weights $(w_j)_{j \geq 0}$ with square-summable first difference, define the discrete trend signal

$$S_t = \sum_{j \geq 0} w_j r_{t-j}.$$

The series is understood as an L^2 limit; square summability and white covariance make it well defined. The same interpretation justifies the reindexing below and passage from finite partial sums to the displayed variance identities. Uncorrelatedness gives

$$\text{Var}(S_t) = \sum_{j \geq 0} w_j^2.$$

Set $w_{-1} = 0$ (and extend a finitely supported kernel by zero past its endpoint). A reindexing of S_{t-1} gives

$$S_t - S_{t-1} = \sum_{j \geq 0} (w_j - w_{j-1}) r_{t-j},$$

and therefore

$$\text{Var}(S_t - S_{t-1}) = \sum_{j \geq 0} (w_j - w_{j-1})^2.$$

To compare kernels at the same signal risk, let

$$P_t = \gamma S_t, \quad \gamma = \frac{\sigma_{\text{sig}}}{(\sum_j w_j^2)^{1/2}},$$

so that $\text{Var}(P_t) = \sigma_{\text{sig}}^2$. The one-period position change $\Delta P_t = P_t - P_{t-1}$ is centered Gaussian with variance

$$\text{Var}(\Delta P_t) = \sigma_{\text{sig}}^2 \frac{\sum_j (w_j - w_{j-1})^2}{\sum_j w_j^2}.$$

For $Z \sim N(0, \tau^2)$, one has $\mathbb{E}|Z| = \sqrt{2/\pi} \tau$. Hence expected absolute turnover is

$$\mathbb{E}|\Delta P_t| = \sigma_{\text{sig}} \sqrt{\frac{2}{\pi}} \left(\frac{\sum_{j \geq 0} (w_j - w_{j-1})^2}{\sum_{j \geq 0} w_j^2} \right)^{1/2}. \quad (16)$$

Any convention that reports one-way turnover as one half of the absolute position change only alters the prefactor. Since the square root is strictly increasing and every other factor in (16) is fixed, minimizing expected turnover is exactly equivalent to minimizing the discrete Rayleigh quotient

$$\mathcal{R}_d(w) := \frac{\sum_j (w_j - w_{j-1})^2}{\sum_j w_j^2}.$$

Thus fixed-risk rescaling is already captured by the denominator, and γ does not need to be carried as an additional optimization variable.

The discrete-to-continuum result provides liminf, recovery, value convergence, and convergence of interpolated near-minimizers to the continuum optimizer; see Theorem A.4 in Appendix A.

10.1 EMA2 under first-moment normalization

The continuous-time EMA2= kernel emphasized by Martin [1] has density proportional to $xe^{-\alpha x}$. Under the present normalization

$$\int_0^\infty f(x) dx = 1, \quad \int_0^\infty xf(x) dx = L,$$

it becomes

$$f_{\text{EMA2}}(x) = \frac{4x}{L^2} e^{-2x/L}.$$

Its Rayleigh quotient is

$$\mathcal{R}(f_{\text{EMA2}}) = \frac{4}{L^2}.$$

Under the shared first-moment normalization, the ratio of the present objective to the EMA2= objective is

$$\frac{\mathcal{R}(f_L)}{\mathcal{R}(f_{\text{EMA2}})} = \frac{2.199071934562496}{4} \approx 0.5497679836.$$

Thus the Rayleigh-value ratio is approximately 0.5497679836, and the corresponding Gaussian expected-turnover ratio is its square root, approximately 0.7414634068. Equivalently, the relative differences are 45.02% and 25.85%, respectively. These numbers compare the kernels after a common normalization; the underlying variational constraints remain different.

The support geometry is also different. The EMA2= kernel has infinite support, whereas the optimizer derived here satisfies $f_L(x) = 0$ for $x > K_L$, where

$$K_L = \kappa_* L, \quad \kappa_* \approx 2.40748835.$$

11 Extensions and scope

The exact finite-support projection-scale result gives the same minimizers for strictly increasing aggregate costs, while turnover-isotropic inputs give exact invariance for absolute turnover; see Propositions B.2, B.4, and B.5. Correlated elliptical inputs lead instead to the generalized quotient (30) of Proposition B.6, and absolutely summable short-memory covariance has the same continuum functional profilewise, as in Proposition B.7. A uniform kurtosis bound gives the stated approximation guarantee, as in Proposition B.8, and the diffuse-kernel CLT gives asymptotic Gaussian turnover under its stated diffuseness and finite-third-moment assumptions.

These conclusions are finite-cutoff, finite-row, or profilewise as applicable; they do not assert a general infinite-process result or convergence of correlated minimizers, and the tilted-Gaussian example in the appendix is deliberately nonstationary. The complete definitions, examples, propositions, proofs, and qualifications appear in Appendix B.

12 Conclusion

This problem is a constrained Dirichlet Rayleigh quotient with a positivity obstacle. The direct method gives existence, the obstacle system and compression argument yield the single compact support interval, and scalar free-boundary analysis identifies the unique optimizer as linear plus trigonometric on $[0, K_L]$, with

$$K_L = \kappa_* L, \\ \kappa_* \approx 2.407488350075723,$$

and

$$L^2 \mathcal{R}(f_L) = 2.199071934562496.$$

The universal constant has an intrinsic scalar-root characterization, with no required elementary closed form, and determines the minimum turnover intensity through $L^2 \mathcal{R}(f_L) = t_*^2 / \kappa_*^2$.

The discrete-to-continuum theorem gives convergence of the scaled minima and of interpolated near-minimizers. The finite-support, distributional, correlated, and discrete conclusions retain the qualifications summarized in Section 11 and proved in Appendices A–B. For reference, the EMA2= kernel studied in [1] has value $4/L^2$ under the same first-moment normalization, while arising from a different variational constraint.

A Discrete-to-continuum convergence

We now make the discrete-to-continuum passage precise. For $h > 0$, let $\mathcal{A}_{h,L}$ denote the class of nonnegative sequences $w = (w_j)_{j \geq 0}$ satisfying

$$w \in \ell^1 \cap \ell^2, \quad Dw \in \ell^2, \quad \sum_{j \geq 0} w_j = 1, \quad h \sum_{j \geq 0} j w_j = L,$$

where $w_{-1} = 0$ and $(Dw)_j = w_j - w_{j-1}$. Define

$$\mathcal{R}_{d,h}(w) := \frac{\|Dw\|_2^2}{\|w\|_2^2}.$$

The numerator and denominator are finite on this class, and the denominator is positive because the weights have unit mass. The natural continuum scaling is $h^{-2} \mathcal{R}_{d,h}$.

For a function v on $[-h, \infty)$ with nonzero square norm, write

$$\widehat{\mathcal{R}}_h(v) := \frac{\int_{-h}^{\infty} |v'(x)|^2 dx}{\int_{-h}^{\infty} v(x)^2 dx}.$$

This auxiliary quotient is taken over the full interpolation domain; it is distinct from $\mathcal{R}$, whose domain is $(0, \infty)$.

Lemma A.1 (Piecewise-linear interpolation identities). *For $w \in \mathcal{A}_{h,L}$, let $I_h w$ be the continuous piecewise-linear function on $[-h, \infty)$ defined by*

$$I_h w(-h) = 0, \quad I_h w(jh) = \frac{w_j}{h}, \quad j \geq 0,$$

and, for general w , define $I_h w$ by linear interpolation on every mesh interval $[jh, (j+1)h]$, using the nodal values w_j/h . Since $w_j \rightarrow 0$, this interpolation has limit zero at infinity. Then $I_h w \geq 0$ and

$$\int_{-h}^{\infty} I_h w(x) dx = 1, \quad (17)$$

$$\int_{-h}^{\infty} x I_h w(x) dx = L, \quad (18)$$

$$\int_{-h}^{\infty} |(I_h w)'(x)|^2 dx = \frac{1}{h^3} \sum_{j \geq 0} (w_j - w_{j-1})^2. \quad (19)$$

Moreover, if $S_h = \sum_j w_j^2$ and $N_h = \sum_j (w_j - w_{j-1})^2$, then

$$\int_{-h}^{\infty} I_h w(x)^2 dx = \frac{S_h}{h} - \frac{N_h}{6h} = \frac{S_h}{h} \left(1 - \frac{\mathcal{R}_{d,h}(w)}{6} \right). \quad (20)$$

Consequently,

$$\widehat{\mathcal{R}}_h(I_h w) = \frac{h^{-2} \mathcal{R}_{d,h}(w)}{1 - \mathcal{R}_{d,h}(w)/6}. \quad (21)$$

Proof. The function $I_h w$ is the linear interpolant through the nodal values w_j/h , with a zero node at $-h$. The hat function associated with the node jh has area h and first moment jh^2 ; summing these contributions gives (17) and (18). On every mesh interval, the derivative is constant, so (19) follows directly. For the L^2 norm, integration of the square of a linear function gives

$$3h \int I_h w^2 = 2S_h + \sum_{j \geq 0} w_j w_{j+1}.$$

Since $w \in \ell^2$ and $w_j \rightarrow 0$, summation by parts gives

$$N_h = 2S_h - 2 \sum_{j \geq 0} w_j w_{j+1},$$

we obtain (20), and then (21). $\square$

Lemma A.2 (Discrete recovery sequence). *Let f_L be the continuum optimizer from Theorem 2.1. There exists, for all sufficiently small $h > 0$, a sequence $w^{(h)} \in \mathcal{A}_{h,L}$ such that*

$$h^{-2} \mathcal{R}_{d,h}(w^{(h)}) \longrightarrow \mathcal{R}(f_L).$$

Proof. Sample f_L on the grid by $v_j^{(h)} = h f_L(jh)$. Since f_L is compactly supported, is C^2 on $[0, K_L]$, and satisfies $f_L(0) = f_L(K_L) = f_L'(K_L) = 0$, the function on $[0, \infty)$ obtained by setting f_L equal to zero past K_L is $C^{1,1}$ and has a bounded weak second derivative; a jump in the classical second derivative at K_L is harmless. The composite trapezoidal estimate for $W^{2,\infty}$ functions, applied also to $x f_L(x)$, therefore gives

$$\sum_j v_j^{(h)} = 1 + O(h^2), \quad h \sum_j j v_j^{(h)} = L + O(h^2).$$

The difference-quotient and Riemann-sum limits also give $h^{-2}\mathcal{R}_{d,h}(v^{(h)}) \rightarrow \mathcal{R}(f_L)$. Choose two nonnegative C_c^2 functions b_1, b_2 supported in a common compact subinterval of $(0, K_L)$ whose mass-moment vectors are linearly independent. Put

$$B_h = \begin{pmatrix} h \sum_j b_1(jh) & h \sum_j b_2(jh) \\ h^2 \sum_j j b_1(jh) & h^2 \sum_j j b_2(jh) \end{pmatrix}, \quad d_h = \begin{pmatrix} 1 - \sum_j v_j^{(h)} \\ L - h \sum_j j v_j^{(h)} \end{pmatrix}.$$

The matrices B_h converge to the nonsingular matrix with columns $(\int b_i, \int x b_i)^T$. They are therefore invertible for all sufficiently small h , with uniformly bounded inverses. The constraint defect of $v^{(h)}$ is $O(h^2)$, so the coefficients $c^{(h)} = B_h^{-1} d_h = O(h^2)$. Defining

$$w_j^{(h)} = v_j^{(h)} + h \sum_{i=1}^2 c_i^{(h)} b_i(jh)$$

then satisfies both discrete constraints exactly. Because f_L is strictly positive on the common compact support of the bumps, the correction preserves nonnegativity for sufficiently small h .

There are $O(h^{-1})$ nonzero corrected entries. The correction has size $O(h^3)$ per entry and first differences of size $O(h^4)$; hence its squared ℓ^2 norm is $O(h^5)$ and the squared norm of its first difference is $O(h^7)$. On the other hand,

$$\|v^{(h)}\|_2^2 = h \int f_L^2 + o(h), \quad \|Dv^{(h)}\|_2^2 = h^3 \int |f_L'|^2 + o(h^3).$$

Cauchy-Schwarz controls the two cross terms: the denominator changes by $O(h^3) = o(h)$ and the numerator by $O(h^5) = o(h^3)$. Therefore $h^{-2}\mathcal{R}_{d,h}(w^{(h)}) \rightarrow \mathcal{R}(f_L)$. $\square$

Lemma A.3 (Discrete limsup). *With $\mathcal{A}_{h,L}$ as above,*

$$\limsup_{h \downarrow 0} \inf_{w \in \mathcal{A}_{h,L}} h^{-2} \mathcal{R}_{d,h}(w) \leq \Lambda_L.$$

Proof. Apply Lemma A.2 to the continuum optimizer f_L . $\square$

Theorem A.4 (Discrete-to-continuum convergence). *Let $h_n \downarrow 0$, and let $w^{(n)} \in \mathcal{A}_{h_n,L}$ be any sequence of ε_n -minimizers, with $\varepsilon_n \rightarrow 0$, in the sense that*

$$h_n^{-2} \mathcal{R}_{d,h_n}(w^{(n)}) \leq \inf_{w \in \mathcal{A}_{h_n,L}} h_n^{-2} \mathcal{R}_{d,h_n}(w) + \varepsilon_n.$$

Let $f_n = I_{h_n} w^{(n)}$. On $[-h_n, 0]$ the interpolant is linear, so

$$\int_{-h_n}^0 f_n = \frac{w_0^{(n)}}{2}, \quad \int_{-h_n}^0 x f_n(x) dx = -\frac{h_n w_0^{(n)}}{6}.$$

The energy bound below implies $w_0^{(n)}/h_n = f_n(0) = O(h_n^{1/2})$, so both quantities tend to zero. The restrictions to $[0, \infty)$ are relatively compact for strong convergence in $L^2(0, \infty)$ and locally uniform convergence on $[0, \infty)$. Every limit f obtained in these two topologies belongs to $\mathcal{A}_L$ and satisfies

$$\mathcal{R}(f) \leq \liminf_{k \rightarrow \infty} h_{n_k}^{-2} \mathcal{R}_{d,h_{n_k}}(w^{(n_k)}).$$

Moreover, if f_L is the unique minimizer from Theorem 2.1, then

$$\lim_{n \rightarrow \infty} \inf_{w \in \mathcal{A}_{h_n,L}} h_n^{-2} \mathcal{R}_{d,h_n}(w) = \mathcal{R}(f_L),$$

and every sequence of discrete near-minimizers satisfies

$$I_{h_n} w^{(n)} \longrightarrow f_L \quad \text{strongly in } L^2(0, \infty) \quad \text{and locally uniformly on } [0, \infty).$$

After zero extension below $-h_n$, the same sequence converges weakly to the zero extension of f_L in $H^1(\mathbb{R})$.

Proof. By Lemma A.3, every sequence of ε_n -minimizers satisfies

$$\limsup_{n \rightarrow \infty} h_n^{-2} \mathcal{R}_{d, h_n}(w^{(n)}) \leq \Lambda_L.$$

Fix an arbitrary subsequence. After first passing to a further subsequence on which the scaled quotients attain their liminf, extract the compact subsequence constructed below and relabel it by n . First note that the discrete difference operator satisfies $N_h \leq 4S_h$, hence $\mathcal{R}_{d, h} \leq 4$. The limsup bound above shows that $h_n^{-2} \mathcal{R}_{d, h_n}(w^{(n)})$ is bounded. By Lemma A.1,

$$\widehat{\mathcal{R}}_h(I_h w) = \frac{h^{-2} \mathcal{R}_{d, h}(w)}{1 - \mathcal{R}_{d, h}(w)/6}$$

is uniformly bounded. Since $\|I_h w\|_1 = 1$, the one-dimensional Gagliardo–Nirenberg inequality therefore gives a uniform bound on $\|I_h w\|_{H^1(-h, \infty)}$. Indeed, writing $E = \|(I_h w)'\|_2^2$ and $D = \|I_h w\|_2^2$, one has $D \leq CE^{1/3}$ and $E \leq CD$, hence E and D are both bounded.

In particular,

$$|I_h w(0)|^2 \leq h \int_{-h}^0 |(I_h w)'(x)|^2 dx \leq hE \longrightarrow 0,$$

so every local uniform limit has trace zero at the origin. Weak compactness in $H^1(\mathbb{R})$, after extending each interpolant by zero on $(-\infty, -h)$, and the one-dimensional compact embedding on bounded intervals give a subsequence converging weakly in $H^1(\mathbb{R})$ and uniformly on compact sets. The positive-half-line first moment is

$$\int_0^\infty x I_h w(x) dx = L + \frac{hw_0}{6}.$$

Consequently, for every $R > 0$, nonnegativity implies the corrected estimate

$$\int_R^\infty I_h w(x) dx \leq \frac{L + hw_0/6}{R}.$$

Lemma 4.2, applied to the translated interpolants, gives a uniformly vanishing L^2 tail as $R \rightarrow \infty$. Thus the convergence is strong in $L^2(0, \infty)$, and the mass constraint passes to the limit. Fatou's lemma also gives

$$m := \int_0^\infty x f(x) dx \leq L.$$

As in the existence proof, the fact that f has mass one implies $m > 0$. The same argument shows that $\mathcal{R}(f) > 0$: otherwise $f' = 0$ almost everywhere, and the L^2 function f would be a vanishing constant, contradicting its unit mass. The negative interpolation strip also vanishes in L^2 : linearity there and the estimate at the origin give

$$\int_{-h}^0 (I_h w)^2 dx = \frac{h}{3} |I_h w(0)|^2 \leq \frac{h^2}{3} E \longrightarrow 0.$$

Hence the zero extensions in fact converge strongly in $L^2(\mathbb{R})$. Weak lower semicontinuity of the full-line Dirichlet energy and strong convergence of the L^2 norm give

$$\mathcal{R}(f) \leq \liminf_n \widehat{\mathcal{R}}_{h_n}(I_{h_n} w^{(n)}).$$

Indeed, the zero extensions converge weakly in $H^1(\mathbb{R})$ and their limit vanishes on $(-\infty, 0)$, so its full-line quotient is exactly $\mathcal{R}(f)$. Since $\mathcal{R}_{d,h_n}(w^{(n)}) = O(h_n^2)$, (21) implies

$$\widehat{\mathcal{R}}_{h_n}(I_{h_n} w^{(n)}) = h_n^{-2} \mathcal{R}_{d,h_n}(w^{(n)})(1 + o(1)).$$

In particular,

$$\mathcal{R}(f) \leq \liminf_n h_n^{-2} \mathcal{R}_{d,h_n}(w^{(n)}).$$

If $m < L$, the dilation

$$\tilde{f}(x) = \frac{m}{L} f\left(\frac{m}{L}x\right)$$

belongs to $\mathcal{A}_L$ and satisfies $\mathcal{R}(\tilde{f}) = (m/L)^2 \mathcal{R}(f) < \mathcal{R}(f)$. But

$$\mathcal{R}(f) \leq \liminf_n h_n^{-2} \mathcal{R}_{d,h_n}(w^{(n)}) \leq \limsup_n h_n^{-2} \mathcal{R}_{d,h_n}(w^{(n)}) \leq \Lambda_L,$$

while $\tilde{f} \in \mathcal{A}_L$ implies $\mathcal{R}(\tilde{f}) \geq \Lambda_L$, a contradiction. Hence $m = L$ and $f \in \mathcal{A}_L$. This also proves the liminf statement.

Since the initial subsequence was arbitrary, the preceding argument proves relative compactness and the asserted liminf inequality for every subsequential limit. Together with Lemma A.3, it gives

$$\lim_{n \rightarrow \infty} \inf_{w \in \mathcal{A}_{h_n, L}} h_n^{-2} \mathcal{R}_{d,h_n}(w) = \Lambda_L.$$

Every subsequential limit of discrete near-minimizers is therefore a minimizer of $\mathcal{R}$ on $\mathcal{A}_L$. Uniqueness in Theorem 2.1 forces the limit to be f_L . Relative compactness then implies convergence of the full sequence: otherwise a subsequence staying a fixed positive distance from f_L in one of the asserted strong topologies would have a further subsequence converging to f_L . Weak $H^1(\mathbb{R})$ convergence of the full zero-extended sequence follows as well: every subsequence has a further weakly convergent subsequence, and the argument above identifies every such weak limit with the zero extension of f_L . $\square$

This theorem gives the liminf, recovery, value-convergence, and near-minimizer conclusions needed to identify the compactly supported continuum kernel as the asymptotic shape on increasingly fine half-line grids. No attainment assertion for a fixed infinite half-line grid is needed here.

B Distributional and dependence extensions

The derivation in Section 10 separates the two probabilistic facts that are needed: a quadratic identity for signal risk and a first absolute moment identity for turnover. This permits an exact characterization that is strictly broader than Gaussianity.

Unless explicitly stated otherwise, the exact non-Gaussian results in this section concern finitely supported coefficient vectors and kernels. This is the natural scope of the projection assumptions below and avoids silently passing distributional identities to infinite random series. The opening Gaussian calculation, by contrast, also covers square-summable kernels through L^2 convergence.

For a finitely supported real sequence $a = (a_j)_{j \geq 0}$, write

$$X_t(a) = \sum_{j \geq 0} a_j r_{t-j}, \quad \|a\|_2^2 = \sum_{j \geq 0} a_j^2.$$

Definition B.1 (Projection-scale white input). A centered input sequence is projection-scale white if there exist $\sigma_r > 0$ and a centered random variable Z , with $\mathbb{E}Z^2 = 1$, such that for every finitely supported a and every t ,

$$X_t(a) \stackrel{d}{=} \sigma_r \|a\|_2 Z. \quad (22)$$

Taking second moments in (22) gives

$$\mathbb{E}[X_t(a)^2] = \sigma_r^2 \|a\|_2^2.$$

By polarization, the returns are homoskedastic with variance σ_r^2 and uncorrelated across time. The converse is false: white covariance fixes second moments but not the distribution of arbitrary linear projections.

Proposition B.2 (General aggregate-cost invariance). *Assume Definition B.1. Let $c : [0, \infty) \rightarrow [0, \infty)$ be Borel and nondecreasing, with the expectations below finite, and let w be a nonzero, finitely supported kernel. Define the cost of the net aggregate trade by*

$$C_t = c(|P_t - P_{t-1}|).$$

After normalizing signal variance to σ_{sig}^2 ,

$$\mathbb{E}C_t = \Psi_c(\mathcal{R}_d(w)), \quad \Psi_c(q) := \mathbb{E}[c(\sigma_{\text{sig}}\sqrt{q} |Z|)]. \quad (23)$$

The function Ψ_c is nondecreasing. Hence every minimizer of $\mathcal{R}_d$ minimizes expected cost. If c is strictly increasing and $\mathbb{P}(|Z| > 0) > 0$, then Ψ_c is strictly increasing and the two problems have exactly the same minimizers.

Proof. Equation (22) gives $\text{Var}(S_t) = \sigma_r^2 \|w\|_2^2$, so fixed signal risk requires

$$\gamma = \frac{\sigma_{\text{sig}}}{\sigma_r \|w\|_2}.$$

Since $S_t - S_{t-1} = X_t(Dw)$,

$$\Delta P_t \stackrel{d}{=} \sigma_{\text{sig}} \frac{\|Dw\|_2}{\|w\|_2} Z = \sigma_{\text{sig}} \sqrt{\mathcal{R}_d(w)} Z.$$

Taking expected cost proves (23). Monotonicity is pointwise in $|Z|$; it is strict under the final two assumptions. $\square$

Thus, on any fixed feasible class of finitely supported kernels, every strictly increasing function of the magnitude of the net trade has the same minimizing kernels under projection-scale white inputs. In particular, this covers linear spread cost, quadratic cost, and superlinear power costs. For one fixed cost exponent, however, the full distributional identity (22) is stronger than necessary; only the corresponding absolute projection moment is needed, as stated below.

Definition B.3 (Turnover-isotropic white input). A centered input sequence is turnover-isotropic white if constants $\sigma_r > 0$ and $m_r > 0$ exist such that, for every finitely supported a and every t :

$$\mathbb{E}[X_t(a)^2] = \sigma_r^2 \|a\|_2^2, \quad \mathbb{E}|X_t(a)| = m_r \|a\|_2. \quad (24)$$

The second identity says exactly that the first absolute projection functional $a \mapsto \mathbb{E}|X_t(a)|$ is a multiple of the Euclidean norm. At the level of objectives formed from finitely supported linear filters, this condition says precisely that expected absolute turnover is a fixed multiple of its root-mean-square value for every coefficient vector, but it does not force spherical finite-dimensional laws. The following explicit process shows the distinction. Under a reference probability $\mathbb{P}_0$, let $(G_j)_{j \in \mathbb{Z}}$ be i.i.d. standard Gaussian, put

$$h(x) = \sin x - \frac{e^{3/2}}{2} \sin(2x),$$

and, for $0 < \varepsilon < (1 + e^{3/2}/2)^{-1}$, define a new probability by

$$\frac{d\mathbb{P}}{d\mathbb{P}_0} = 1 + \varepsilon h(G_0).$$

This density is positive and integrates to one because h is bounded and odd. Set $r_j = G_j$, now viewed under $\mathbb{P}$. The identity

$$\mathbb{E}_0[Z \sin(cZ)] = ce^{-c^2/2}, \quad Z \sim N(0, 1),$$

shows, with $c = 1, 2$, that $\mathbb{E}_0[G_0 h(G_0)] = 0$. This centers r_0 ; for $j \neq 0$, independence and $\mathbb{E}_0 h(G_0) = 0$ center r_j as well. Moreover, every integrable functional H of finitely many coordinates that is invariant under simultaneous sign reversal has the same expectation under $\mathbb{P}$ and $\mathbb{P}_0$: its correction term $\mathbb{E}_0[Hh(G_0)]$ vanishes by global sign reversal. Taking $H = r_i r_j$ and $H = |X_t(a)|$ gives, for every finitely supported a and every t ,

$$\mathbb{E}[r_i r_j] = \delta_{ij}, \quad \mathbb{E}|X_t(a)| = \sqrt{\frac{2}{\pi}} \|a\|_2.$$

Thus this is a turnover-isotropic white input. Nevertheless every block containing time zero is nonspherical: its density relative to the corresponding Gaussian block is not even invariant under simultaneous sign reversal, because h is nonzero and odd. The example is deliberately nonstationary; stationarity is not part of Definition B.3. It proves that the two projection-moment identities use strictly less distributional information than spherical symmetry.

Indeed, if a is finitely supported and $\sum_j a_j = 0$, then

$$w_j = \sum_{k=0}^j a_k$$

is finitely supported and satisfies $Dw = a$. Conversely, zero-sum finite sequences are dense in ℓ^2 . To see this, let a be supported in $\{0, \dots, N\}$ and put $s = \sum_j a_j$. For $M \geq 1$, append M remote coefficients, each equal to $-s/M$. The resulting sequence $a^{(M)}$ has sum zero and

$$\|a^{(M)} - a\|_2 = \frac{|s|}{\sqrt{M}} \longrightarrow 0.$$

Under the first identity in (24), the map $a \mapsto \mathbb{E}|X_t(a)|$ is σ_r -Lipschitz in ℓ^2 , because

$$|\mathbb{E}|X_t(a)| - \mathbb{E}|X_t(b)|| \leq \mathbb{E}|X_t(a - b)| \leq (\mathbb{E}X_t(a - b)^2)^{1/2} = \sigma_r \|a - b\|_2.$$

Thus requiring the expected-turnover identity for every finite kernel forces the second identity in (24) by continuity.

Proposition B.4 (Exact non-Gaussian invariance). *For a turnover-isotropic white input and a nonzero, finitely supported kernel, fixed-risk expected absolute turnover is*

$$\mathbb{E}|\Delta P_t| = \frac{m_r}{\sigma_r} \sigma_{\text{sig}} \sqrt{\mathcal{R}_d(w)}. \quad (25)$$

Consequently, on every common feasible class of nonzero, finitely supported kernels, the discrete variance and expected absolute turnover problems have exactly the same minimizers. Under the continuum scaling of Section 10, their common profile-dependent objective is the variational problem solved in Theorem 2.1; the white-noise half-line-grid asymptotics are covered by Theorem A.4.

Proof. By (24), $\text{Var}(S_t) = \sigma_r^2 \|w\|_2^2$, so fixed signal risk requires $\gamma = \sigma_{\text{sig}}/(\sigma_r \|w\|_2)$. Since $S_t - S_{t-1} = X_t(Dw)$, the second identity in (24) gives

$$\mathbb{E}|\Delta P_t| = \gamma m_r \|Dw\|_2 = \frac{m_r}{\sigma_r} \sigma_{\text{sig}} \frac{\|Dw\|_2}{\|w\|_2}.$$

This is (25). $\square$

The same invariance holds for every power of the aggregate trade, provided linear projections form a common scale family at that moment order.

Proposition B.5 (Power-law aggregate costs). *Fix $g \geq 1$ and suppose that, for every finitely supported a and every t ,*

$$\mathbb{E}|X_t(a)|^g = m_{r,g} \|a\|_2^g \quad (26)$$

for a constant $m_{r,g} > 0$, while $\mathbb{E}[X_t(a)^2] = \sigma_r^2 \|a\|_2^2$. If the per-period cost is

$$C_t = \eta |P_t - P_{t-1}|^g, \quad \eta > 0,$$

then, for every nonzero, finitely supported kernel, after normalizing signal variance to σ_{sig}^2 ,

$$\mathbb{E}C_t = \eta \frac{m_{r,g}}{\sigma_r^g} \sigma_{\text{sig}}^g \mathcal{R}_d(w)^{g/2}. \quad (27)$$

Consequently, on any fixed discrete feasible class of finitely supported kernels, for every exponent g for which the stated moment identity holds, the minimizer set is the Rayleigh minimizer set. In particular, whenever the identity holds for several exponents, those exponents all give the same minimizers. At the continuum level the profile-dependent objective is a positive constant times $\mathcal{R}(f)^{g/2}$, whose unique minimizer is the profile in Theorem 2.1.

For white Gaussian inputs,

$$\frac{m_{r,g}}{\sigma_r^g} = \mathbb{E}|Z|^g = \frac{2^{g/2} \Gamma((g+1)/2)}{\sqrt{\pi}}, \quad Z \sim N(0, 1). \quad (28)$$

Thus the conclusion applies in particular to $g = 1$, $g = 3/2$, and $g = 2$.

Proof. Fixed signal risk gives $\gamma = \sigma_{\text{sig}}/(\sigma_r \|w\|_2)$. Since $\Delta P_t = \gamma X_t(Dw)$, equation (26) gives

$$\mathbb{E}C_t = \eta \gamma^g m_{r,g} \|Dw\|_2^g,$$

which is (27). The map $x \mapsto x^{g/2}$ is strictly increasing on $[0, \infty)$, so its minimizers are exactly those of $\mathcal{R}_d$. Finally, for a standard normal Z , the substitution $u = x^2/2$ gives

$$\mathbb{E}|Z|^g = \frac{2}{\sqrt{2\pi}} \int_0^\infty x^g e^{-x^2/2} dx = \frac{2^{g/2} \Gamma((g+1)/2)}{\sqrt{\pi}},$$

proving (28). $\square$

For a specified exponent g , condition (26) directly guarantees, at the level of all finitely supported linear projections, that the power-cost objective is a fixed increasing function of $\mathcal{R}_d(w)$. By contrast, Definition B.1 is a simple distributional condition that guarantees the conclusion simultaneously for all exponents and, more generally, for every increasing aggregate cost covered by Proposition B.2.

The projection-scale class contains white Gaussian inputs, but also genuinely dependent non-Gaussian examples. If Z_j are independent standard Gaussians and $V \geq 0$ is independent of the whole sequence, with $\mathbb{E}V = 1$, then

$$r_j = \sqrt{V} Z_j$$

is uncorrelated but dependent whenever V is nonconstant. For every a , $X_t(a)/\|a\|_2$ has the common distribution $\sqrt{V}G$, where $G \sim N(0, 1)$ is independent of V . Thus it satisfies Definition B.1, and hence also Definition B.3, with $\sigma_r = 1$ and $m_r = \sqrt{2/\pi} \mathbb{E}\sqrt{V}$: indeed, conditioning on V gives

$$X_t(a) \mid V \sim N(0, V\|a\|_2^2).$$

More generally, a process whose finite-dimensional blocks are centered spherical with scalar covariance is projection-scale when the standardized one-dimensional projection law is common across horizons. Matching only the first absolute projections of such a process is already sufficient for turnover isotropy.

Uncorrelatedness alone is not sufficient. For independent Rademacher inputs $\varepsilon_1, \varepsilon_2 \in \{-1, 1\}$,

$$\mathbb{E}|a\varepsilon_1 + b\varepsilon_2| = \frac{|a+b| + |a-b|}{2} = \max\{|a|, |b|\},$$

which is not proportional to $(a^2 + b^2)^{1/2}$.

B.1 Correlated elliptical inputs

There is also an exact correlated formulation. Let Γ be the covariance operator of the return sequence and assume that, for every finitely supported a and every t ,

$$\mathbb{E}|X_t(a)| = c_{\text{ell}} \langle a, \Gamma a \rangle^{1/2}. \quad (29)$$

On each fixed finite horizon, this projection identity holds for a centered elliptically contoured input vector whenever its first absolute moment exists. It holds with one constant across all horizons when the standardized one-dimensional projection law is common; see Cambanis, Huang, and Simons [15].

Proposition B.6 (Generalized Rayleigh quotient). *Under (29), fixed-risk expected absolute turnover for a finitely supported kernel satisfying $\langle w, \Gamma w \rangle > 0$ is*

$$\mathbb{E}|\Delta P_t| = c_{\text{ell}} \sigma_{\text{sig}} \left(\frac{\langle Dw, \Gamma Dw \rangle}{\langle w, \Gamma w \rangle} \right)^{1/2}. \quad (30)$$

If Γ is a scalar multiple of the identity, this reduces exactly to the problem solved above. For a general covariance operator it is a different, generalized Rayleigh problem.

Proof. Fixed signal risk gives $\gamma = \sigma_{\text{sig}} / \langle w, \Gamma w \rangle^{1/2}$. Apply (29) to Dw . $\square$

For a stationary input with spectral density $s(\omega)$ and the transfer function $W(\omega) = \sum_{j \geq 0} w_j e^{-ij\omega}$, the squared quotient in (30) is, by the standard spectral representation of

stationary covariance [14],

$$\mathcal{R}_s(w) = \frac{\int_{-\pi}^{\pi} 4 \sin^2(\omega/2) |W(\omega)|^2 s(\omega) d\omega}{\int_{-\pi}^{\pi} |W(\omega)|^2 s(\omega) d\omega}. \quad (31)$$

Indeed, the denominator is the spectral covariance identity for w , while first differencing multiplies $W(\omega)$ by $1 - e^{-i\omega}$, whose squared modulus is $4 \sin^2(\omega/2)$. Thus correlation generally changes the finite-scale optimizer by weighting frequencies differently.

B.2 Short-memory universality

Although the finite-scale correlated problem differs, its long-lookback limit is unchanged for a broad short-memory class.

Proposition B.7 (Continuum limit under short memory). *Let (r_t) be stationary with autocovariance $(\gamma_h)_{h \in \mathbb{Z}}$ satisfying*

$$\sum_{h \in \mathbb{Z}} |\gamma_h| < \infty, \quad \Gamma_0 := \sum_{h \in \mathbb{Z}} \gamma_h > 0.$$

Let s be its continuous spectral density and Γ its covariance operator. Let $f \in C_c^1([0, \infty))$ be nonzero, satisfy $f(0) = 0$, and set

$$w_j^{(L)} = \frac{1}{L} f\left(\frac{j}{L}\right), \quad j \geq 0,$$

with zero extension to negative indices. As $L \rightarrow \infty$ through positive integers, one has

$$L \langle w^{(L)}, \Gamma w^{(L)} \rangle \longrightarrow \Gamma_0 \int_0^\infty f^2, \quad (32)$$

$$L^3 \langle Dw^{(L)}, \Gamma Dw^{(L)} \rangle \longrightarrow \Gamma_0 \int_0^\infty |f'|^2. \quad (33)$$

Consequently,

$$L^2 \mathcal{R}_s(w^{(L)}) \longrightarrow \mathcal{R}(f). \quad (34)$$

Proof. Write each covariance quadratic form by lags:

$$\langle u, \Gamma u \rangle = \sum_{h \in \mathbb{Z}} \gamma_h \sum_j u_j u_{j-h}.$$

For each fixed h , the Riemann-sum limits are

$$L \sum_j w_j^{(L)} w_{j-h}^{(L)} \longrightarrow \int f^2$$

and, since $Dw_j^{(L)} = L^{-1}[f(j/L) - f((j-1)/L)]$,

$$L^3 \sum_j Dw_j^{(L)} Dw_{j-h}^{(L)} \longrightarrow \int |f'|^2.$$

Cauchy–Schwarz gives, uniformly in the lag h ,

$$\left| L \sum_j w_j^{(L)} w_{j-h}^{(L)} \right| \leq L \left(\sum_j (w_j^{(L)})^2 \right) = O(1),$$

and, likewise,

$$\left| L^3 \sum_j Dw_j^{(L)} Dw_{j-h}^{(L)} \right| \leq L^3 \sum_j (Dw_j^{(L)})^2 = O(1).$$

The implied constants are independent of h and L because the corresponding zero-lag quantities converge to finite limits. Hence the lagged terms are bounded by a constant multiple of $|\gamma_h|$, which is summable by assumption. Dominated convergence over h therefore proves (32) and (33). Their ratio gives (34). $\square$

This proves profilewise universality of the continuum functional for each fixed smooth compactly supported profile. For correlated elliptical inputs it also gives the same profilewise limiting expected-turnover objective, up to a constant. The discrete convergence theorem above applies to the white-noise Rayleigh problem; the proposition here does not assert convergence or equicoercivity of minimizers for a family of correlated discrete problems.

B.3 Robustness without projection isotropy

Even when expected absolute turnover is not exactly quadratic, a discrete Rayleigh minimizer has a distribution-free approximation guarantee under a uniform kurtosis bound.

Proposition B.8 (Kurtosis robustness). *Assume the inputs are centered and white with unit variance, and suppose that for every finitely supported a ,*

$$\mathbb{E}[X_t(a)^4] \leq K \|a\|_2^4 \quad (35)$$

for some $K \geq 1$. Then every nonzero, finitely supported kernel satisfies

$$\frac{\sigma_{\text{sig}}}{\sqrt{K}} \sqrt{\mathcal{R}_d(w)} \leq \mathbb{E}|\Delta P_t| \leq \sigma_{\text{sig}} \sqrt{\mathcal{R}_d(w)}. \quad (36)$$

If, over the same feasible class of finitely supported kernels, minimizers w_R of $\mathcal{R}_d$ and w_T of exact expected absolute turnover exist, then

$$\mathbb{E}|\Delta P_t(w_R)| \leq \sqrt{K} \mathbb{E}|\Delta P_t(w_T)|. \quad (37)$$

Proof. For any centered X with finite fourth moment, interpolation between L^1 , L^2 , and L^4 gives

$$\mathbb{E}|X| \geq \frac{(\mathbb{E}X^2)^{3/2}}{(\mathbb{E}X^4)^{1/2}},$$

because Hölder’s inequality yields

$$\mathbb{E}X^2 = \mathbb{E}(|X|^{2/3}|X|^{4/3}) \leq (\mathbb{E}|X|)^{2/3}(\mathbb{E}X^4)^{1/3}.$$

On the other hand, Cauchy–Schwarz gives $\mathbb{E}|X| \leq (\mathbb{E}X^2)^{1/2}$. Apply these inequalities to $X = X_t(Dw)$, use (35), and normalize signal variance. This proves (36). Finally,

$$T(w_R) \leq \sigma_{\text{sig}} \sqrt{\mathcal{R}_d(w_R)} \leq \sigma_{\text{sig}} \sqrt{\mathcal{R}_d(w_T)} \leq \sqrt{K} T(w_T),$$

where $T(w) = \mathbb{E}|\Delta P_t(w)|$. $\square$

Proposition B.9 (Diffuse-kernel central limit). *Let $(r_j)_{j \in \mathbb{Z}}$ be i.i.d., centered, have unit variance, and satisfy $\mathbb{E}|r_0|^3 < \infty$. Let $a^{(n)}$ be nonzero finitely supported coefficient sequences such that*

$$\frac{\max_j |a_j^{(n)}|}{\|a^{(n)}\|_2} \rightarrow 0.$$

Then

$$\frac{X_t(a^{(n)})}{\|a^{(n)}\|_2} \Rightarrow N(0, 1) \quad \text{and} \quad \frac{\mathbb{E}|X_t(a^{(n)})|}{\|a^{(n)}\|_2} \rightarrow \sqrt{\frac{2}{\pi}}.$$

Proof. The Lyapunov ratio is bounded by

$$\frac{\sum_j \mathbb{E}|a_j^{(n)} r_j|^3}{\|a^{(n)}\|_2^3} \leq \mathbb{E}|r_0|^3 \frac{\max_j |a_j^{(n)}|}{\|a^{(n)}\|_2} \rightarrow 0.$$

The Lyapunov central limit theorem [11] gives convergence in distribution. The normalized sums all have second moment one, so their absolute values are uniformly integrable; convergence of first absolute moments follows. $\square$

Applying Proposition B.9 to $a^{(n)} = Dw^{(n)}$ shows that Gaussian expected turnover is asymptotically valid for long kernels satisfying $\max_j |Dw_j^{(n)}|/\|Dw^{(n)}\|_2 \rightarrow 0$, well beyond the Gaussian class.

C Certified numerical reproducibility

The file `certify_rayleigh_constants.py`, distributed with the Lean 4 formalization of Appendix D, uses Arb ball arithmetic via `python-flint` version 0.9.0. All elementary operations and evaluations of sin and cos are therefore performed with rigorous enclosures rather than ordinary floating-point error estimates. The script declares `python-flint==0.9.0` in its PEP 723 inline metadata, and `uv` creates the isolated environment for it.

The script evaluates Q at the exact decimal rationals

$$t_- = 3.5701290278126360584735350083717233860170522483300$$

and

$$t_+ = 3.5701290278126360584735350083717233860170522483302.$$

At 100 decimal digits of working precision, Arb certifies

$$Q(t_-) \in [-1.553, -1.552] \times 10^{-47}, \quad Q(t_+) \in [1.448, 1.449] \times 10^{-47}.$$

Lemma 7.1 proves that Q has exactly one admissible zero and that Q is strictly increasing on the interval containing these endpoints, so $t_* \in (t_-, t_+)$. The script evaluates the closed forms on the interval hull $[t_-, t_+]$ and propagates the enclosure through forms for A_* , C_* , I_0 , I_1 , κ , λ_* , $F'_*(0)$, and $F''_*(1)$. It certifies both endpoint derivative signs and asserts that every resulting ball lies strictly inside the rounding bin of the decimal printed in the paper. From the root of the released repository, the complete certificate is reproduced by

```
uv run --script manuscript/certify_rayleigh_constants.py
```

D Lean 4 formalization

The development accompanying this paper is a Lean 4 package built on mathlib, published at https://github.com/marchinidavide/rayleigh_kernel. The Lean toolchain version, the mathlib revision, and all transitive dependencies are pinned in that repository, so the checked statements are reproducible from the recorded commits alone.

Theorem 2.1 is recorded as the aggregate structure `MainOptimizerData`, which collects the attainment of Λ_L , the uniqueness of the minimizer, the support and positivity geometry, the explicit profile with its scalar root. It also records the equivalent inequality form displayed above together with its equality case.

The discrete convergence theorem is recorded as `tendsto_scaledDiscreteMinimum` and `tendsto_scaled_rayleighQuotient_of_nearMinimizer`, and the objective ratio against the EMA2= benchmark of Section 10 is recorded in `RayleighKernel/Benchmarks`. The decimal constants printed in this paper are enclosed inside Lean by `tStar_mem_rootBracket` and the accompanying rounding-bin theorems, independently of the Python certificate of Appendix C. The file `RayleighKernel/Claims.lean` gives the full statement-by-statement correspondence, and the Gaussian expected-turnover ratio quoted in Section 10 is not formalized.

The formalization retains the qualifications of Section 11 and of Appendices A–B. The probability and covariance results are finite-cutoff, finite-row, or profilewise statements rather than theorems about an infinite random series or a stationary process; the short-memory limit is stated for each fixed smooth compactly supported profile; no Γ -convergence package is included; and the tilted-Gaussian example of Appendix B is left unformalized.

The release gate shipped with the package compiles the library, rejects `sorry`, `admit`, and `native_decide` in the production sources, and audits the axioms of the public declarations. That audit reports only `propext`, `Classical.choice`, and `Quot.sound`, so the formalized statements rest on no assumption beyond Lean’s standard axiom system. The formalization is a companion to the proofs above rather than a substitute for them, and the two are maintained together.

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Statement on the use of generative AI

The Lean 4 formalization accompanying this paper and the proofs given here were produced with help from `gpt-5.6-sol` on high reasoning effort. The author verified all the results reported here and takes full responsibility for the content of this paper.

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